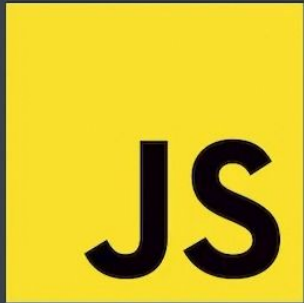


# Javascript

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## console.log( )

To write (log) a message on the console

```
console.log("Apna College");
```

```
console.log(1234);
```

```
console.log(2+2);
```

```
console.log("Apna", "College", 123);
```



New Tab

Search Google or type a URL

APNA COLLEGE

nasa moonjiraSeat Matrix 2022Department of Co...font-awesome - Li...unicodes JS

ElementsConsoleSourcesNetworkPerformanceMemory

topFilterDefault levels3 Issues: 3

> "Hello!"

< 'Hello!'

> console.log("Hello!");

Hello!

VM247:1

< undefined

> console.log("Hello World from Apna College!");

Hello World from Apna College!

VM299:1

< undefined

> let num = 123;

< undefined

> console.log("num");

num

< undefined

> console.log(num);

123

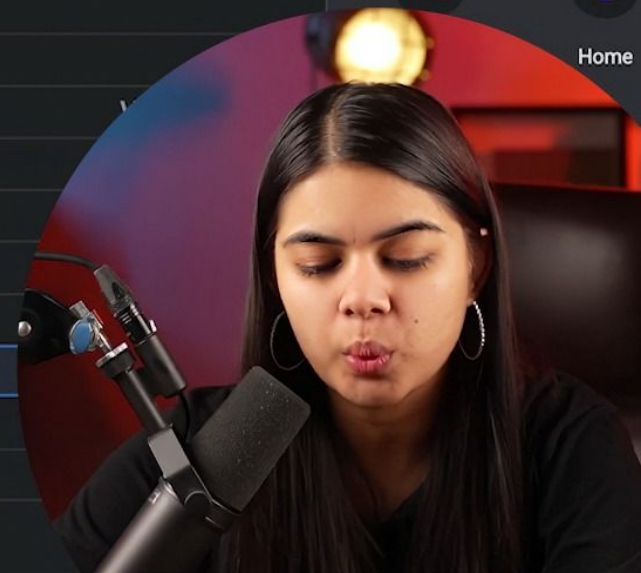
< undefined

>

Goo

Search Google or ty

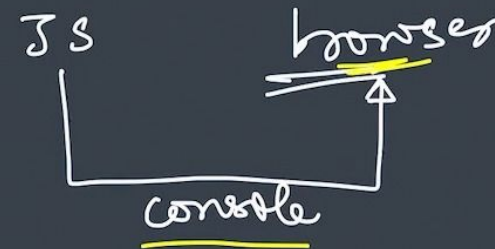
```
Hello! VM247:1
< undefined
> console.log("Hello World from Apna College!");
Hello World from Apna College! VM299:1
< undefined
> let num = 123;
< undefined
> console.log("num");
num VM404:1
< undefined
> console.log(num);
123
< undefined
> console.log(1+2);
3
< undefined
> console.log("Hello", "World", num, (1+5));
Hello World 123 6
< undefined
>
```





## console.log( )

To write (log) a message on the console



```
console.log("Apna College");
```

```
console.log(1234);
```

```
console.log(2+2);
```

```
console.log("Apna", "College", 123);
```



# Linking JS File

```
<script src="app.js"> </script>
```



# Linking JS File

```
<script src="app.js"> </script>
```

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JS





EXPLORER

JS

app.js

index.html 2



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OUTLINE

&lt;&gt; index.html 2 JS app.js

&lt;&gt; index.html &gt; html &gt; head &gt; script

```
1  <!DOCTYPE html>
2  <html lang="en">
3    <head>
4      <meta charset="UTF-8" />
5      <meta http-equiv="X-UA-Compatible" content="IE=edge" />
6      <meta name="viewport" content="width=device-width, initial-scale=1.0" />
7      <title>JavaScript Concepts</title>
8      <script src="app.js">
9    </head>
10   <body>
11     <h1>Hello JavaScript!</h1>
12   </body>
13 </html>
14
```

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EXPLORER

JS

app.js

index.html

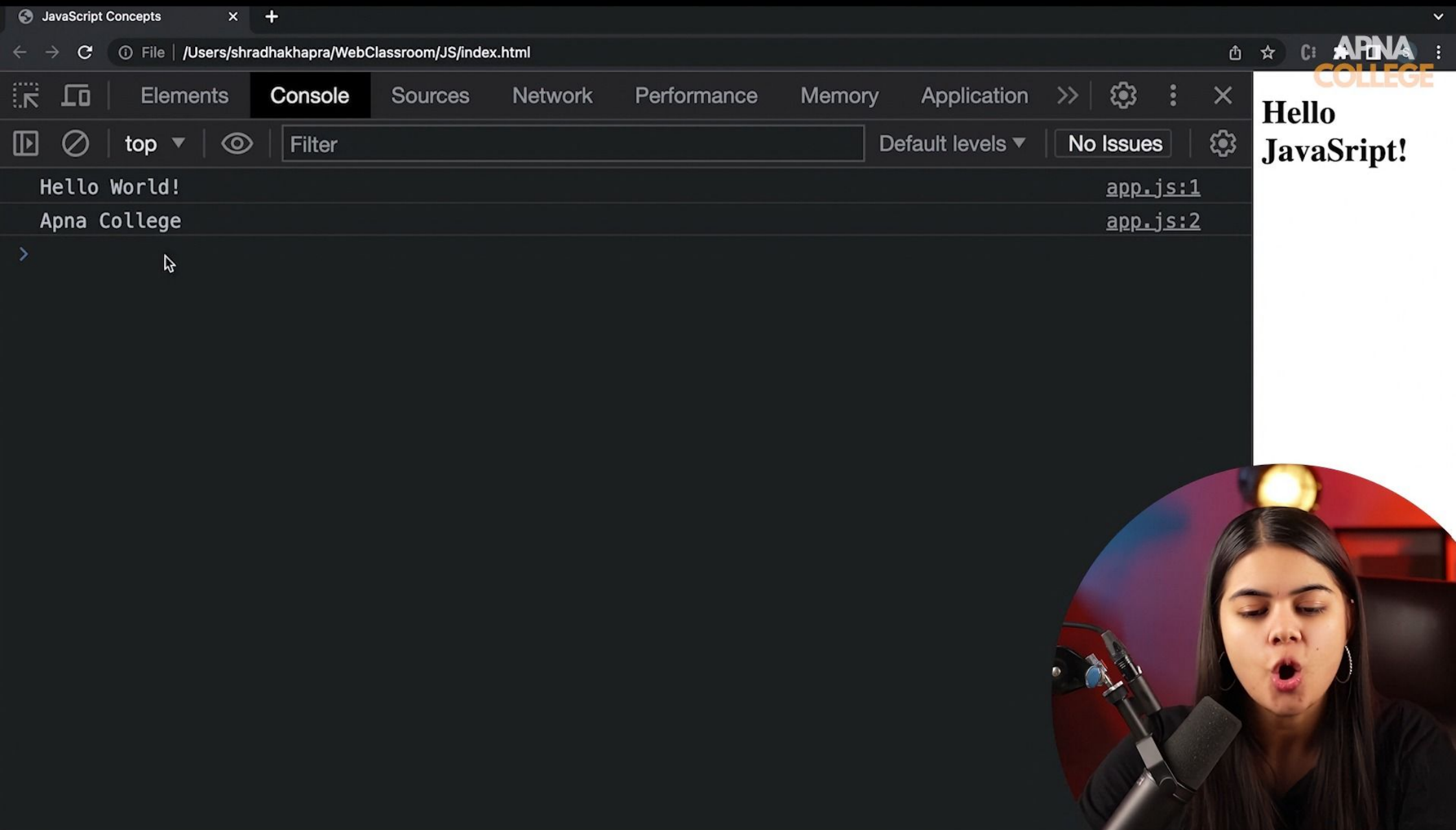
index.html JS app.js

index.html &gt; html &gt; body

```
1 <!DOCTYPE html>
2 <html lang="en">
3   <head>
4     <meta charset="UTF-8" />
5     <meta http-equiv="X-UA-Compatible" content="IE=edge" />
6     <meta name="viewport" content="width=device-width, initial-scale=1.0" />
7     <title>JavaScript Concepts</title>
8
9   </head>
10  <body>
11    <h1>Hello JavaScript!</h1>
12    <script src="app.js"></script>
13  </body>
14 </html>
15
```



OUTLINE



EXPLORER

&lt;&gt; index.html

JS app.js

JS

JS app.js

&lt;&gt; index.html

JS app.js &gt; ...

```
1 console.log("Hello World!");  
2 console.log("Apna College");  
3 let a = 10;  
4 let b = 5;  
5 console.log("sum is : ", a+b)
```

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JavaScript Concepts

File | /Users/shradhakhapra/WebClassroom/JS/index.html

ElementsConsoleSourcesNetworkPerformanceMemoryApplication

topFilterDefault levelsNo Issues

Hello World!app.js:1

Apna Collegeapp.js:2

sum is : 15app.js:5

>

APNA COLLEGE

Hello JavaScript

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06:00 / 08:03



EXPLORER



<> index.html

JS app.js



JS

JS app.js

<> index.html

JS app.js > ...

```
1 // console.log("Hello World!");
2 // console.log("Apna College");
3 let a = 10;
4 let b = 5
5 console.log("sum is : ", a + b);
6 //this is a comment
7
```

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# Template Literals

They are used to add embedded expressions in a string.

```
let a = 5;
```

```
let b = 10;
```

```
console.log(`Your pay ${a + b} rupees`);
```

```
// console.log("Price is", a+b, "rupees");
```



&lt;&gt; index.html

JS app.js

JS app.js &gt; ...

```
1 let pencilPrice = 10;  
2 let eraserPrice = 5;  
3 console.log("the total price is : ", pencilPrice + eraserPrice, " Rupees. ");|
```

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JavaScript Concepts

File | /Users/shradhakhapra/WebClassroom/JS/index.html

ElementsConsoleSourcesNetworkPerformanceMemoryApplication

topFilterDefault levelsNo Issues

The total price is : 15 Rupees. app.js:3

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Hello JavaScript

index.html JS app.js

JS app.js &gt; [x] output

```
1 let pencilPrice = 10;
2 let eraserPrice = 5;
3 let output =
4 | "The total price is : " + (pencilPrice + eraserPrice) + " Rupees. ";
5 console.log(output);
6
```

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&lt;&gt; index.html

JS app.js

JS app.js &gt; [x] output

```
1 let pencilPrice = 10;
2 let eraserPrice = 5;
3 // let output = "The total price is : " + (pencilPrice + eraserPrice) + " Rupees.";
4 let output = `The total price is : ${pencilPrice} Rupees.` //back ticks
5 console.log(output);
6
```





JavaScript Concepts

← → ↻ File | /Users/shradhakhapra/WebClassroom/JS/index.html

Elements

Console

Sources

Network

Performance

Memory

Application

>>

⚙️ ⋮ ✕

⏮ ⏪ ⏩ ⏭

top ▾

👁

Filter

Default levels ▾

No Issues

⚙️

The total price is : 10 Rupees. app.js:5

>

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Hello JavaScript



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# Template Literals

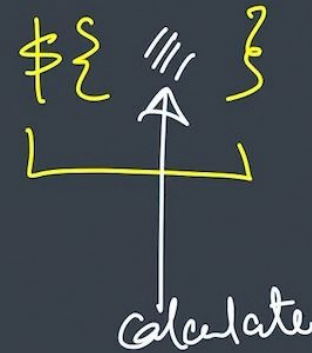
They are used to add embedded expressions in a string.

```
let a = 5;
```

```
let b = 10;
```

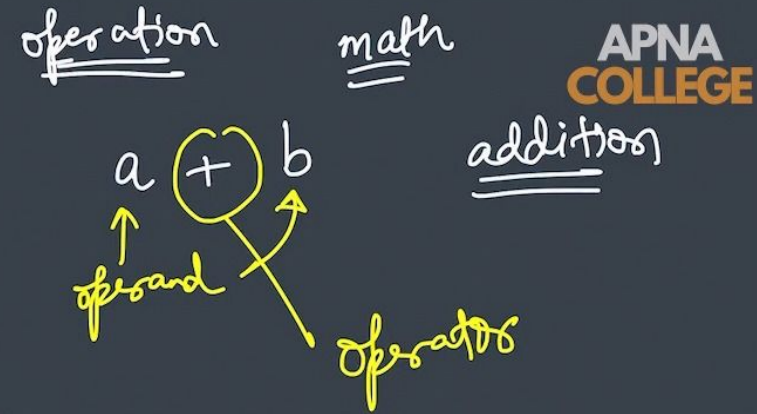
```
console.log(`Your pay ${a + b} rupees`);
```

```
// console.log("Price is", a+b, "rupees");
```



# Operators in JS

- Arithmetic (+, -, \*, /, %, \*\*)
- Unary (++ , --)
- Assignment (=, +=, -=, \*=, /=, %= etc.)
- Comparison
- Logical



$a++$   
 $b+$



# Comparison Operators

Comparison Operators to **compare 2 values**

>

>=

<

<=

==

!=

JS



# Comparison Operators

Comparison Operators to **compare 2 values**

&gt;

&gt;=

&lt;

&lt;=

==

!=

let age = ~~99~~; 16  
age > 18  
↓      ↓  
true   false





# Comparison Operators

Comparison Operators to **compare 2 values**

> → Greater than  
>= → greater than or equal to  
< → lesser than  
<= → lesser than or equal to  
==  
!=

||

let age = 23;

age > 18

true

false

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age >= 18



# Comparison Operators

==

- compares value, not type

```
> "123" == 123
< true
> 1 == '1'
< true
> 0 == ''
< true
> 0 == false
< true
> null == undefined
< true
```

===

- compares type & value

```
> "123" === 123
< false
> 1 === '1'
< false
> 0 === ''
< false
> 0 === false
< false
> null === undefined
< false
```



## Comparison for non-numbers

```
> 'a' > 'A'
```

```
< true
```

'a' -> 61, 'b' -> 62 ...

```
> 'a' > 'b'
```

```
< false
```

'A' -> 41 'B' -> 42 ...

```
> 'b' < 'c'
```

```
< true
```

```
> 'B' < 'C'
```

```
< true
```

```
> '*' < '&'
```

```
< false
```



## Comparison for non-numbers

```
> 'a' > 'A'
```

```
< true
```

```
> 'a' > 'b'
```

```
< false
```

```
> 'b' < 'c'
```

```
< true
```

```
> 'B' < 'C'
```

```
< true
```

```
> '*' < '&'
```

```
< false
```

'a' -> 61, 'b' -> 62 ...

'A' -> 41 'B' -> 42 ...

value → unique val  
↓  
unicode

'a' - 'z'

'A' - 'Z'

\$, %, \*







0000

C0 Controls and Basic Latin

007F

|   | 000                            | 001                            | 002                              | 003                          | 004                          | 005                          | 006                          | 007                          |
|---|--------------------------------|--------------------------------|----------------------------------|------------------------------|------------------------------|------------------------------|------------------------------|------------------------------|
| 0 | <div>NUL</div> <div>0000</div> | <div>DLE</div> <div>0010</div> | <div>SP</div> <div>0020</div>    | <div>0</div> <div>0030</div> | <div>@</div> <div>0040</div> | <div>P</div> <div>0050</div> | <div>`</div> <div>0060</div> | <div>p</div> <div>0070</div> |
| 1 | <div>SOH</div> <div>0001</div> | <div>DC1</div> <div>0011</div> | <div>!</div> <div>0021</div>     | <div>1</div> <div>0031</div> | <div>A</div> <div>0041</div> | <div>Q</div> <div>0051</div> | <div>a</div> <div>0061</div> | <div>q</div> <div>0071</div> |
| 2 | <div>STX</div> <div>0002</div> | <div>DC2</div> <div>0012</div> | <div>"</div> <div>0022</div>     | <div>2</div> <div>0032</div> | <div>B</div> <div>0042</div> | <div>R</div> <div>0052</div> | <div>b</div> <div>0062</div> | <div>r</div> <div>0072</div> |
| 3 | <div>ETX</div> <div>0003</div> | <div>DC3</div> <div>0013</div> | <div>#</div> <div>0023</div>     | <div>3</div> <div>0033</div> | <div>C</div> <div>0043</div> | <div>S</div> <div>0053</div> | <div>c</div> <div>0063</div> | <div>s</div> <div>0073</div> |
| 4 | <div>EOT</div> <div>0004</div> | <div>DC4</div> <div>0014</div> | <div>\$</div> <div>0024</div>    | <div>4</div> <div>0034</div> | <div>D</div> <div>0044</div> | <div>T</div> <div>0054</div> | <div>t</div> <div>0064</div> | <div>t</div> <div>0074</div> |
| 5 | <div>ENQ</div> <div>0005</div> | <div>NAK</div> <div>0015</div> | <div>%</div> <div>0025</div>     | <div>5</div> <div>0035</div> | <div>E</div> <div>0045</div> | <div>U</div> <div>0055</div> | <div>u</div> <div>0065</div> | <div>u</div> <div>0075</div> |
| 6 | <div>ACK</div> <div>0006</div> | <div>SYN</div> <div>0016</div> | <div>&amp;</div> <div>0026</div> | <div>6</div> <div>0036</div> | <div>F</div> <div>0046</div> | <div>V</div> <div>0056</div> | <div>v</div> <div>0066</div> | <div>v</div> <div>0076</div> |





## Comparison for non-numbers

```
> 'a' > 'A'  
< true
```

```
> 'a' > 'b'  
< false
```

```
> 'b' < 'c'  
< true
```

```
> 'B' < 'C'  
< true
```

```
> '*' < '&'  
< false
```

'a' → 61, 'b' → 62 ...

'A' → 41 'B' → 42 ...

lowercase

'a' - 'z'

a < b < c < d < e < ...

A < B < C < D < ...

value → unique val

↓  
unicode

'a' - 'z'

'A' - 'Z'

\$, %, \*, &



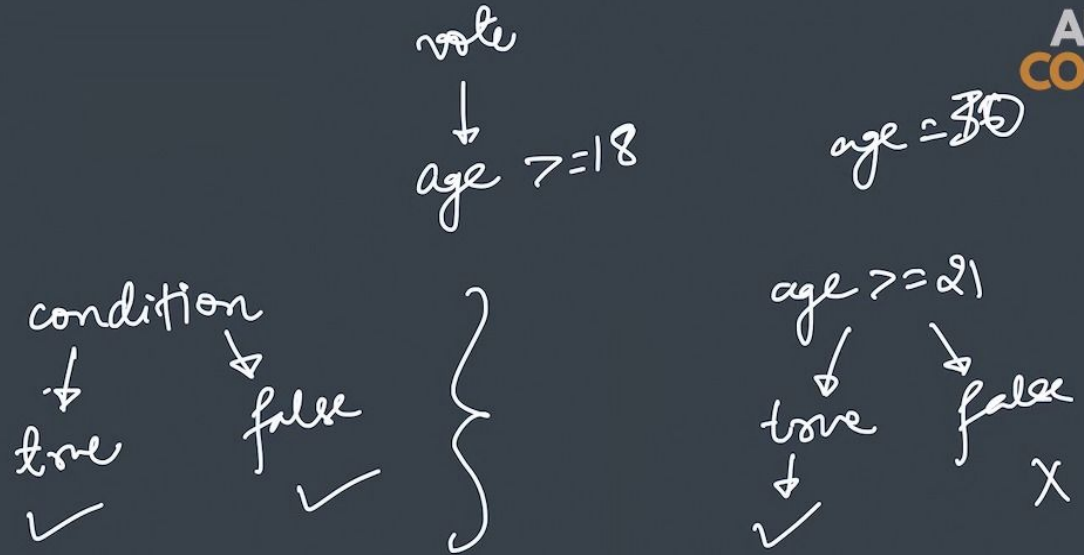
# Conditional Statements

- if-else
- nested if-else
- switch



# Conditional Statements

- if-else
- nested if-else
- switch



# if Statement

// some code before if

```
if (some condition) {  
  // DO SOMETHING  
}
```

// some code after if

let age = 19

age >= 18

↓  
false

↓  
true



&lt;&gt; index.html

JS app.js

JS app.js &gt; ...

```
1 //Conditional Statements
2 console.log("before my if statement");
3 let age = 14;
4 if (age >= 18) {
5   console.log("you can vote");
6   console.log("you can drive");
7 }
8 console.log("after my if statement");
9
```





&lt;&gt; index.html

JS app.js X

JS app.js &gt; ...

```
1 //Conditional Statements
2 console.log("Before if statement");
3 let age = 25;
4 if (age >= 18) {
5   console.log("you can vote");
6   console.log("you can drive");
7   let a = 5;
8   console.log(5 * a);
9 }
10 console.log("after my if statement");
11
```

[See Real World Examples From GitHub](#)

(method) Console.log(...data: any[]): void



&lt;&gt; index.html

JS app.js X

JS app.js &gt; ...

```
1  //Conditional Statements
2  console.log("before my if statement");
3  let age = 23;
4  if (age >= 18) {
5      console.log("you can vote");
6  }
7  (age < 18) {
8      console.log("you cannot vote");
9  }
10 console.log("after my if statement");
11
```

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05:31 / 08:09



&lt;&gt; index.html

JS app.js X

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JS app.js &gt; ...

```
1  //Conditional Statements
2  console.log("before my if statement");
3  let age = 14;
4  if (age >= 18) {
5      console.log("you can vote");
6  }
7  if (age < 18) {
8      console.log("you cannot vote");
9  }
10 console.log("after my if statement");
11
```



&lt;&gt; index.html

JS app.js

JS app.js &gt; [🔗] firstName

```
1  💡 Conditional Statements
2  let firstName = "shradha";
3  if (firstName == "shradha") {
4    console.log(`Welcome ${firstName}`);
5  }
6
```



# Practice Qs

Qs. Create a traffic light system that shows what to do based on color.



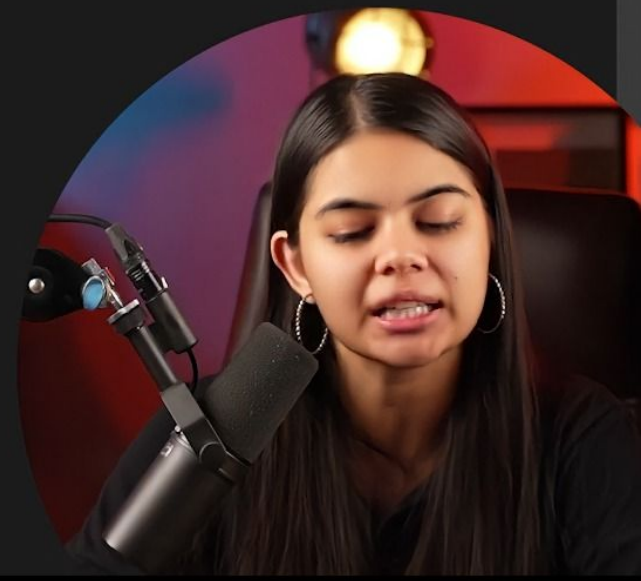


JS app.js > ...

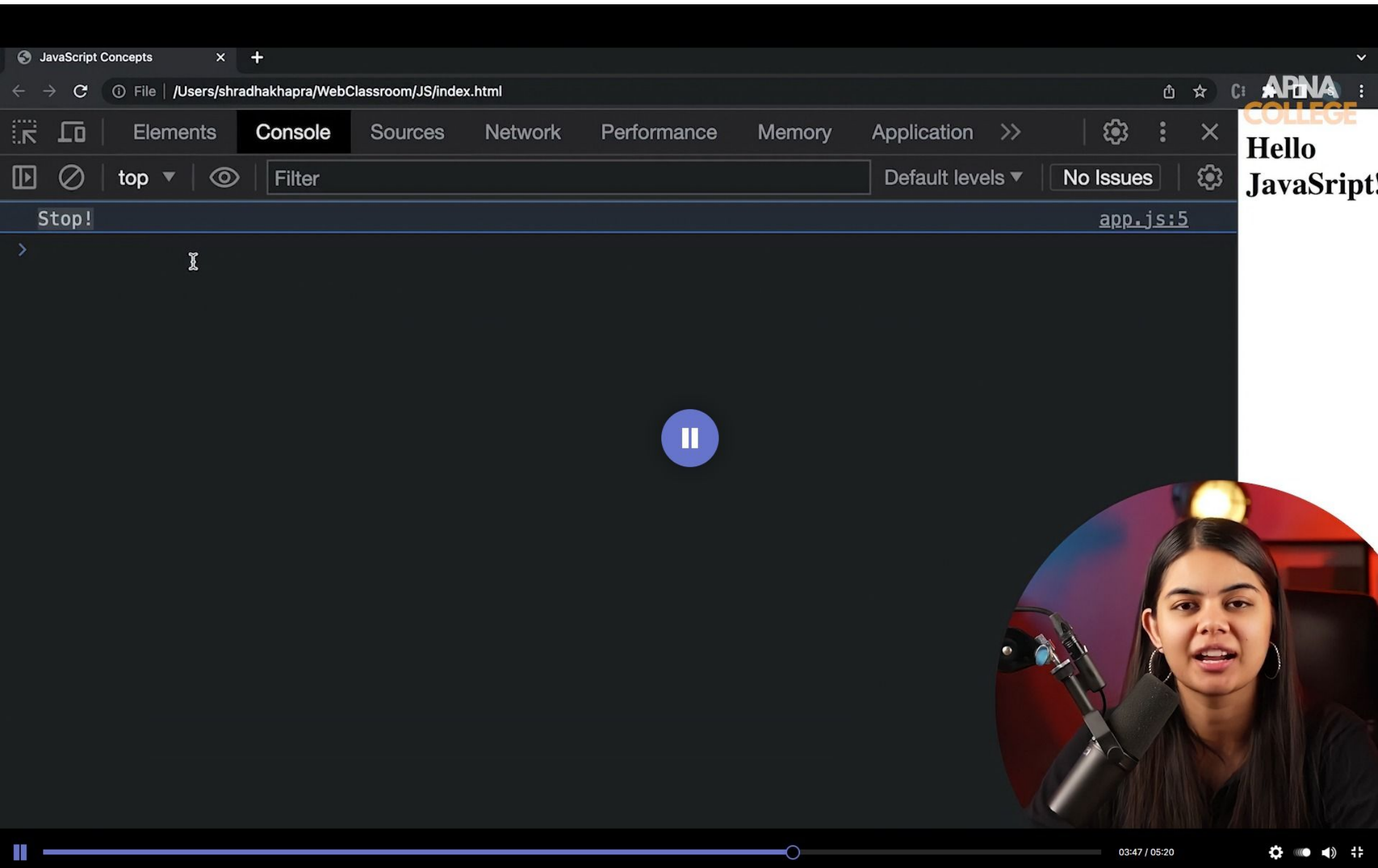
```

1  let color = "red";
2
3  //Traffic Light System
4  if (color === "red") {
5      console.log("Stop!");
6  }
7  (color === "yellow") {
8      console.log("slow down");
9  }
10 if (color === "green") {
11     console.log("go");
12 }
13

```



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# else if Statement

```
if (condition1) {  
    // DO SOMETHING  
}  
  
else if (condition2) {  
    // DO SOMETHING ELSE  
}  
  
else if (condition3) {  
    // DO SOMETHING ELSE  
}
```



&lt;&gt; index.html

JS app.js X

JS app.js &gt; [age]

```
1  // let color = "green";
2
3  // //Traffic Light System
4  // if (color === "red") {
5  //   console.log("Stop! light color is red");
6  // }
7  // if (color === "yellow") {
8  //   console.log("Slow down. light color is yellow");
9  // }
10 // if (color === "green") {
11 //   console.log("Go. light color is green");
12 // }
13
14 let age = 14;
15 if (age >= 18) {
16   console.log("you can vote");
17 } else if (age < 18) {
18   console.log("you cannot vote");
19 }
20
```

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# else Statement

```
if (condition1) {
```

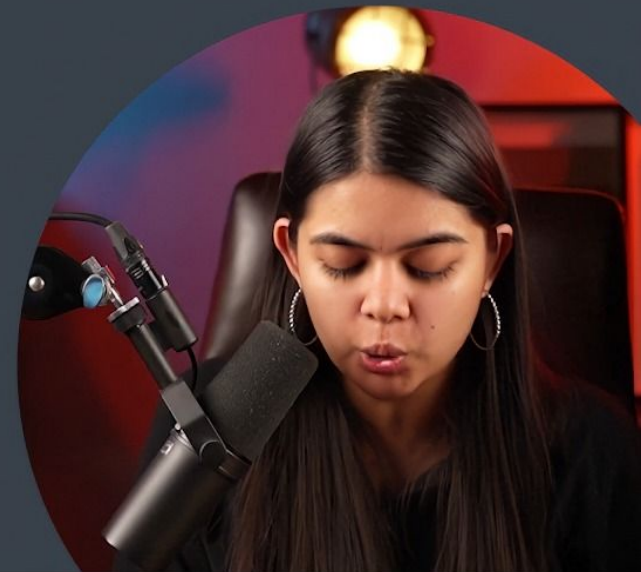
```
  // DO SOMETHING
```

```
}
```

```
else {
```

```
  // DO SOMETHING ELSE
```

```
}
```





&lt;&gt; index.html

JS app.js



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JS app.js &gt; ...

```
1  let age = 17;
2  if (age >= 18) {
3    console.log("you can vote");
4  } else {
5    console.log("you cannot vote");
6  }
7
8  let color = "white";
9
10 if (color === "red") {
11   console.log("stop");
12 } else if (color === "yellow") {
13   console.log("slow down");
14 } else if (color === "green") {
15   console.log("go");
16 } else {
17   console.log("traffic light is broken");
18 }
19
```



## Practice Qs



Qs. Create a system to calculate popcorn prices based on the size customer asked for :

if size is 'XL', price is Rs. 250

if size is 'L', price is Rs. 200

if size is 'M', price is Rs. 100

if size is 'S', price is Rs. 50



&lt;&gt; index.html

JS app.js



JS app.js &gt; ...

```
1  let size = "XL";
2
3  if (size === "XL") {
4    console.log("price is Rs. 250");
5  } else if (size === "L") {
6    console.log("price is Rs. 200");
7  } else if (size === "M") {
8    console.log("price is Rs. 100");
9  } else {
10   console.log("price is Rs. 50");
11 }
12
```

log(...data: any[]): void

<> index.html

JS app.js

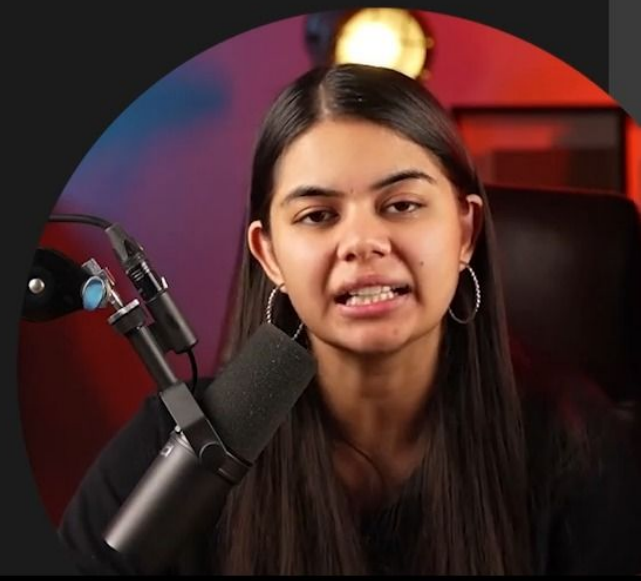


JS app.js > ...

```

1  let marks = 32;
2
3  if (marks >= 33) {
4      console.log("Pass");
5      if (marks >= 80) {
6          console.log("Grade : 0");
7      } else {
8          console.log("Grade : A");
9      }
10 } else {
11     console.log("Better luck next time!");
12 }
13

```



# Logical Operators

Logical Operators to **combine expressions**

**&&** Logical AND

(exp1) && (exp2)

```
> true && true  
< true  
> true && false  
< false  
> false && true  
< false  
> false && false  
< false
```

$a + b$





<> index.html

JS app.js



JS app.js > ...

```

1 //Logical Operators
2 let marks = 75;
3 
4 if (marks >= 33 && marks >= 80) {
5     console.log("pass");
6     console.log("A+");
7 }
8

```



&lt;&gt; index.html

JS app.js

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JS app.js &gt; ...

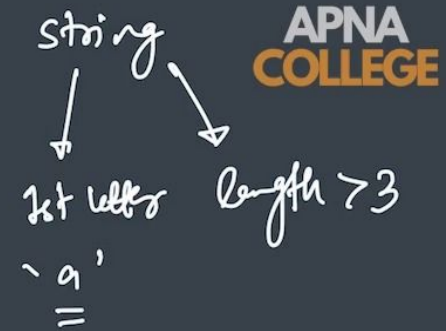
```
1 //Logical Operators
2 let marks = 75;
3 
4 if ((marks > 33 && marks <= 80) || !false) {
5   console.log("pass");
6 }
7
```

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## Practice Qs

Qs. A "good string" is a string that starts with the letter 'a' & has a length > 3. Write a Program to find if a string is good or not.



Qs. Predict the output of following code :

```
let num = 12;

if((num%3 === 0) && ( (num+1 == 15) || (num-1 == 11) ) ) {
  console.log("safe");
} else {
  console.log("unsafe");
}
```



## Practice Qs

let str = "a b b l e"  
          0 1 2 3 4

string  
↓      ↓  
1st letter    length > 3  
= 'a'

Qs. A "good string" is a string that starts with the letter 'a' & has a length > 3. Write a Program to find if a string is good or not.

str[0] === 'a'      str.length

Qs. Predict the output of following code :

```
let num = 12;  
  
if((num%3 === 0) && ( (num+1 == 15) || (num-1 == 11) ) ) {  
  console.log("safe");  
} else {  
  console.log("unsafe");  
}
```





## Practice Qs

let str = "a b b l e"  
 0 1 2 3 4

string  
 ↓  
 1st letter = 'a'  
 length > 3

Qs. A "good string" is a string that starts with the letter 'a' & has a length > 3. Write a Program to find if a string is good or not.

if ( (str[0] === 'a') && (str.length > 3) )  
 {  
 log("Good");  
 }

Qs. Predict the output of following code :

3 else

```
let num = 12;

if((num%3 === 0) && ( (num+1 == 15) || (num-1 == 11) ) ) {
  console.log("safe");
} else {
  console.log("unsafe");
}
```





## Practice Qs

Qs. A "good string" is a string that starts with the letter 'a' & has a length > 3. Write a Program to find if a string is good or not.

$$12 \div 3 = 4$$

$$3 \overline{)12} \quad 0$$

$$\text{num} = \underline{12}$$

$$12 + 1 = 15?$$

$$13 \neq 15$$

$$12 - 1 = 11?$$

$$11 = 11 \checkmark$$

if (true)

Qs. Predict the output of following code :

```
let num = 12;
if( (num%3 === 0) && ( (num+1 == 15) || (num-1 == 11) ) ) {
  console.log("safe");
} else {
  console.log("unsafe");
}
```

Handwritten annotations on the code:

- A bracket above the entire if condition is labeled 'T' (True).
- A bracket above the first condition `(num%3 === 0)` is labeled 'T' (True).
- A bracket above the second condition `(num+1 == 15)` is labeled 'F' (False).
- A bracket above the third condition `(num-1 == 11)` is labeled 'T' (True).



## Practice Qs

Qs. A "good string" is a string that starts with the letter 'a' & has a length > 3. Write a Program to find if a string is good or not.

$$\text{num} = \underline{12}$$

if (true)

Qs. Predict the output of following code :

```
let num = 12;  
if( (num%3 === 0) && ( (num+1 == 15) || (num-1 == 11) ) ) {  
  console.log("safe");  
} else {  
  console.log("unsafe");  
}
```

$$12 \div 3 = 0$$
$$3 \overline{)12} 0$$

$$12+1 = 15?$$

$$13 \neq 15$$

$$12-1 = 11?$$

$$11 = 11 \checkmark$$

"safe"



# truthy & falsy

Everything in JS is true or false (in boolean context).

This doesn't mean their value itself is false or true, but they are treated as false or true if taken in boolean context.

## Falsy values

false, 0, -0, 0n (BigInt value), "" (empty string), null, undefined, NaN

## Truthy values

Everything else



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JS app.js



JS app.js &gt; ...

```
1  let string = " ";
2
3  if (string) {
4    console.log("string is not empty");
5  } else {
6    console.log("string is empty");
7  }
8
9  let num = -10;
10
11  if (num) {
12    console.log("num is not equal to 0");
13  } else {
14    console.log("num is equal to 0");
15  }
16
```





# Switch Statement

Used when we have some fixed values that we need to compare to.

```
let color = "red";

switch(color) {
  case "red" :
    console.log("stop");
    break;
  case "yellow" :
    console.log("slow down");
    break;
  case "green" :
    console.log("GO");
    break;
  default :
    console.log("Broken Light");
}
```





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JS app.js



JS app.js &gt; ...

```
1  let color = "red";
2
3  switch (color) {
4      case "red":
5          console.log("stop");
6          break;
7      case "yellow":
8          console.log("slow down");
9          break;
10     case "green":
11         console.log("go");
12         break;
13     default:
14         console.log("light is broken");
15 }
16
```



## Practice Qs



Qs. Use switch statement to print the day of the week using a number variable 'day' with values 1 to 7.

1 = Monday, 2 = Tuesday & so on

let day = 1 / 2 / 3 .... 7



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JS app.js



JS app.js &gt; [?] day

```
1  let day = 7;
2
3  switch (day) {
4    case 1:
5      console.log("Monday");
6      break;
7    case 2:
8      console.log("Tuesday");
9      break;
10   case 3:
11     console.log("Wednesday");
12     break;
13   case 4:
14     console.log("Thursday");
15     break;
16   case 5:
17     console.log("Friday");
18     break;
19   case 6:
20     console.log("Saturday");
21     break;
22   case 7:
23     console.log("Sunday, fun day");
24     break;
25   default:
26     console.log("wrong day!");
27 }
28
```



03:10 / 04:25



# Alert & Prompt

**Alert** displays an alert message on the page.

```
alert("something is wrong!");
```

**Prompt** displays a dialog box that asks user for some input.

```
prompt("please enter your roll no.");
```





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JS app.js

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JS app.js

```
1 alert("this is a simple alert!");
2 console.log("this is a simple log");
3 console.error("this is an error msg");
4
```



02:19 / 09:08





No Issues

```
this is a simple log app.js:1  
✖ this is an error msg app.js:2  
  (anonymous) @ app.js:2
```

# Hello JavaScript!

APNA  
COLLEGE

02:54 / 09:08



&lt;&gt; index.html

JS app.js

JS app.js

```
1 console.log("this is a simple log");  
2 console.error("this is an error msg");  
3 console.warn("this is a warning msg");  
4
```



JavaScript Concepts

File | /Users/shradhakhapra/WebClassroom/JS/index.html

Elements Console

No Issues

this is a simple log

app.js:1

✖ this is an error msg

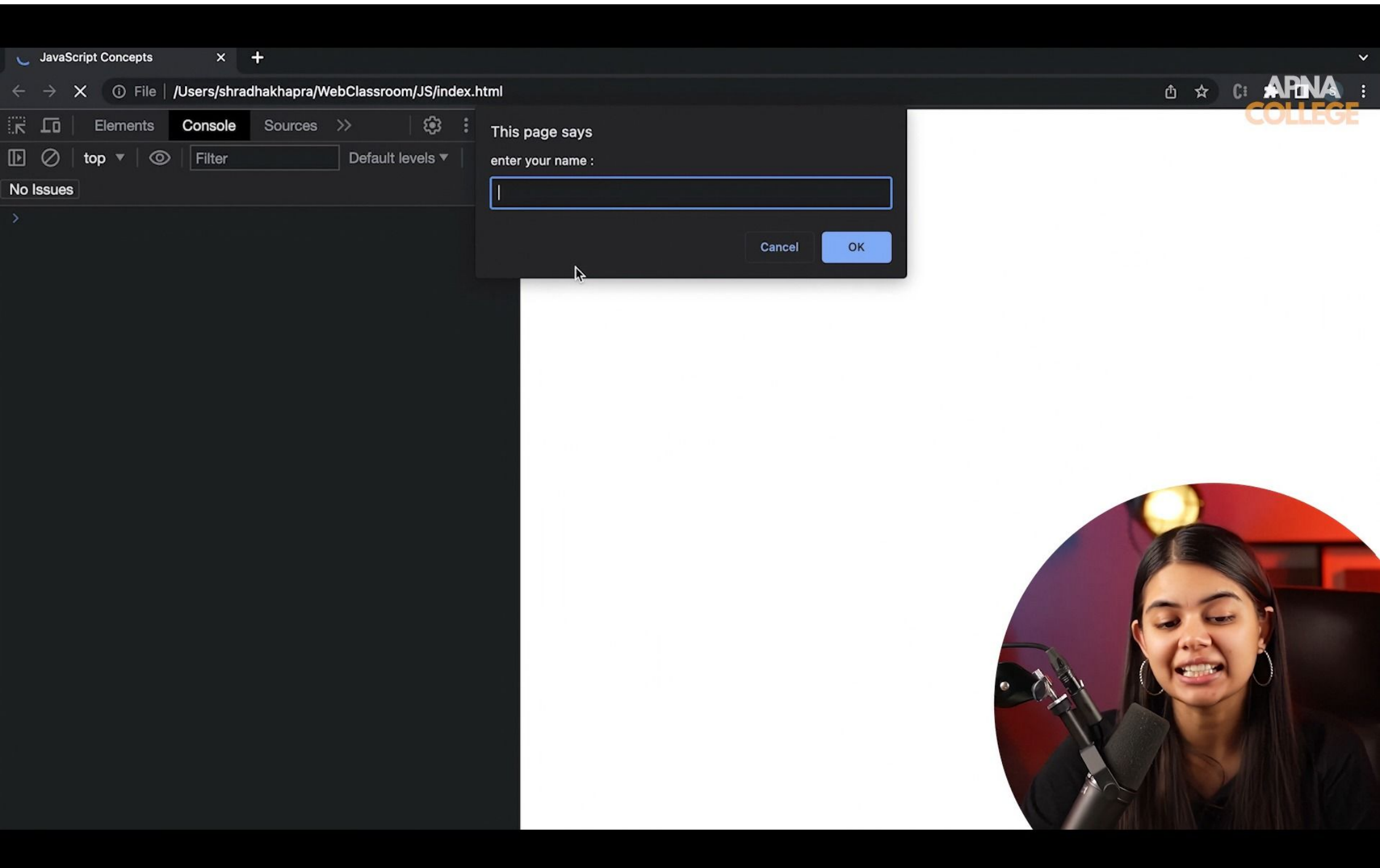
app.js:2

⚠ this is a warning msg

app.js:3

# Hello JavaScript!

amerkhan082630@gmail.com



index.html JS app.js

JS app.js &gt; firstName

```
1 let firstName = prompt("enter your name : ");
2 console.log(firstName);
3
```





## JS (Part 2)

### Practice Questions

Qs1. Create a number variable num with some value.  
Now, print "good" if the number is divisible by 10 and print "bad" if it is not.

Qs2. Take the user's name & age as input using prompts.  
Then return back the following statement to the user as an alert (by substituting their name & age) :  
**name is age years old.**

[Use template Literals to print this sentence]

Qs3. Write a switch statement to print the months in a quarter.  
Months in Quarter 1 : January, February, March  
Months in Quarter 2 : April, May, June  
Months in Quarter 3 : July, August, September  
Months in Quarter 4: October, November, December

[Use the number as the case value in switch]

Qs4. A string is a **golden string** if it starts with the character 'A' or 'a' and has a total length greater than 5.  
For a given string print if it is golden or not.

Qs5. Write a program to find the largest of 3 numbers.

Qs6 (Bonus). Write a program to check if 2 numbers have the same last digit.  
Eg : 32 and 47852 have the same last digit i.e. 2